Combination of Electrical Resistivity Tomography, Ground Penetrating Radar, Geotechnical Data & GIS
for Dam Stability Assessment in a Flood Risk Mitigation Perspective

Study Area: Abidjan, Côte d'Ivoire — A Comprehensive Beginner-Friendly ArcGIS Pro Guide

---

Project Goals

- bullet Integrate and spatially model geophysical (ERT & GPR) and geotechnical datasets inside a GIS environment.
- bullet Combine subsurface investigation results with topographic, hydrological, and land-use data.
- bullet Perform multi-criteria spatial analysis (AHP-based Weighted Overlay) for dam vulnerability assessment.
- bullet Produce decision-support maps for flood risk management and infrastructure resilience planning.

Study Area — Abidjan, Côte d'Ivoire

Abidjan is located in the south of Côte d'Ivoire between latitudes 5°10′N and 5°38′N and longitudes 3°40′W and 4°10′W. The district encompasses 13 communes, sits on a lagoon/coastal sedimentary basin, and faces recurrent flooding due to rapid urbanisation, low-lying topography, and heavy rainfall (1,400–2,400 mm/yr).

Small earth dams and retention basins (barrages) exist within the Abidjan district to manage run-off (e.g., barrage de la Djibi, barrage de la Mé, barrage d'Adzopé). Their structural health directly affects downstream flood risk. This project assesses their stability by combining geophysical imaging of the dam body with spatial factors.

Software & Prerequisites

- bullet **ArcGIS Pro 3.x** with *Spatial Analyst*, *3D Analyst*, and *Geostatistical Analyst* extensions.
- bullet **RES2DINV / EarthImager** or similar for ERT inversion (produces inverted resistivity model).
- bullet **ReflexW / GPR-SLICE / GPRPy** for GPR data processing.
- bullet Basic familiarity with GIS concepts (layers, projections, attribute tables).
- bullet A working internet connection for data downloads.

> If you have never opened ArcGIS Pro before, complete the free Esri training "*Getting Started with ArcGIS Pro*" (~4 hours) at `learn.arcgis.com` first.

Coordinate Reference System Convention

All data in this project will be stored and analysed in:

WGS 1984 / UTM Zone 30N (EPSG:32630)

Abidjan falls into UTM Zone 30N. Using a projected CRS ensures that distances (buffer, proximity) are in metres.

PHASE 1 — Data Acquisition & Sourcing

This section lists every dataset you need, where to get it, and the format you should expect.

1.1 Administrative Boundaries (Abidjan District & Communes)

1. Go to **data.humdata.org** and search "*Côte d'Ivoire administrative boundaries*". Download the shapefile ZIP labelled `civ_admbnda_adm2_cntig_2020.zip` (Admin Level 2 = communes).
2. Alternative: **geoboundaries.org** → search "Côte d'Ivoire" → ADM1 and ADM2.
3. Alternative: **GADM** (gadm.org) → download the GeoPackage or shapefile for CIV.

> You need two layers: one for the entire Abidjan district boundary (for clipping all other data), and one for the commune polygons (for per-commune summaries on the dashboard).

1.2 Digital Elevation Model (DEM)

1. Go to **earthexplorer.usgs.gov**. Create a free account.
2. Under *Search Criteria*, draw a bounding box around Abidjan (roughly 5.1–5.7°N, 3.7–4.2°W).
3. Under *Data Sets*, expand **Digital Elevation › SRTM › SRTM 1 Arc-Second Global**.
4. Add the tiles to your cart and download the GeoTIFF(s).

> Tiles covering Abidjan: typically `n05_w004_1arc_v3.tif` and possibly `n05_w005_1arc_v3.tif`. Resolution = 30 m.

Alternative (12.5 m resolution): ALOS PALSAR DEM from the Alaska Satellite Facility (`search.asf.alaska.edu`). Higher resolution but requires more processing.

1.3 Rainfall / Precipitation Data

1. Go to **data.chc.ucsb.edu/products/CHIRPS-2.0/**.
2. Download monthly GeoTIFFs for Côte d'Ivoire, or the Africa-wide set and clip later. Choose the last 10–30 years for mean annual rainfall.
3. Alternative: **WorldClim** (worldclim.org/data/worldclim21.html) provides 30-year climate normals (BIO12 = annual precipitation) at 1 km resolution. Easier for beginners.

1.4 Land Use / Land Cover (LULC)

1. **BNETD 2020 Côte d'Ivoire Land Cover** (10 m, Sentinel-2 based). Available on Google Earth Engine (`BNETD/land_cover/v1`). Export the Abidjan clip as GeoTIFF.
2. **ESA WorldCover 2021** (10 m global). Download from `worldcover2021.esa.int`.
3. Alternative: **ESA CCI Land Cover** (300 m, 1992–2020, good for change detection). Download from the Copernicus Climate Data Store.

1.5 Soil Data

1. **FAO Harmonized World Soil Database v2.0** (`gaez.fao.org/pages/hwsd`). Download the raster + attribute database. Fields include texture class, drainage class, gravel content.
2. **ISRIC SoilGrids 250m** (`soilgrids.org`). Download clay %, sand %, bulk density tiles for the Abidjan area.
3. These give regional context; your own geotechnical boreholes override them locally.

1.6 Hydrology & Drainage Network

1 **HydroSHEDS** (`hydrosheds.org`) provides river network lines and catchment polygons derived from SRTM. Download HydroRIVERS (lines) for Africa.

2 OpenStreetMap waterways: download from **data.humdata.org** “Côte d’Ivoire waterways” or use **QuickOSM** inside ArcGIS Pro.

3 For dam-specific catchments you will delineate watersheds from the DEM yourself (Step 3 in the workflow).

1.7 Infrastructure / Roads / Buildings

1 **OpenStreetMap via HDX**: search “Côte d’Ivoire OpenStreetMap buildings / roads” at data.humdata.org.

2 Or use **QuickOSM** plugin inside ArcGIS Pro to query buildings, roads, and bridges around dam sites.

1.8 Dam / Barrage Locations

1 **GRanD (Global Reservoir and Dam Database)** – download from `globaldamwatch.org/grand`. Filter to CIV.

2 Supplement with **local knowledge**: Ministry of Hydraulics (MINEF), Office National de l’Eau Potable (ONEP), or direct GPS surveys of small barrages.

3 Create a CSV (Latitude, Longitude, Name, Type, Height_m, Year_Built) and you will import it as a point feature class.

1.9 Geophysical Field Data (ERT & GPR)

These come from **your own fieldwork** at each dam site. Below is what to prepare:

Electrical Resistivity Tomography (ERT)

bullet Conduct ERT surveys along the dam crest and toe using a multi-electrode system (e.g., Wenner or dipole-dipole array).

bullet Invert raw data in **RES2DINV** or **EarthImager 2D**.

bullet Export the **inverted model** (X, Depth/Elevation, Resistivity) as a CSV or XYZ file. Each row = one model cell centroid.

bullet Also record the profile GPS start/end coordinates (RTK or handheld GPS) so you can georeference the profile.

Ground Penetrating Radar (GPR)

bullet Run GPR lines along and across the dam body (200–400 MHz antenna for civil structures).

bullet Process in **ReflexW** or **GPR-SLICE**: dewow, background removal, gain, migration, depth conversion.

bullet Export **depth-slice amplitude maps** (horizontal slices at regular depth intervals) as georeferenced rasters (GeoTIFF). Alternatively export as XYZ point cloud.

Geotechnical Borehole Data

bullet Compile borehole logs into a spreadsheet: Borehole_ID, X, Y, Depth_from, Depth_to, Lithology, SPT_N, Moisture_%, Permeability.

bullet Save as CSV. You will import this as a point feature class and optionally create subsurface cross-sections.

> If you do not yet have field data, you can proceed with synthetic/demo data to learn the GIS workflow, then swap in real measurements later.

Data Inventory Summary

#	Dataset	Source	Format	Resolution
1	Admin boundaries (ADM2)	HDX / GADM	Shapefile / GPKG	Vector
2	DEM (SRTM 1 arc-sec)	USGS EarthExplorer	GeoTIFF	30 m
3	Rainfall (CHIRPS / WorldClim)	UCSB / WorldClim	GeoTIFF	~5 km / 1 km
4	Land Use / Land Cover	BNETD 2020 / ESA WorldCover	GeoTIFF	10 m
5	Soil (HWSD / SoilGrids)	FAO / ISRIC	Raster + DBF	1 km / 250 m
6	Rivers / drainage	HydroSHEDS / OSM	Shapefile	Vector
7	Roads / buildings	OpenStreetMap (HDX)	Shapefile	Vector
8	Dam locations	GRanD + field GPS	CSV → Points	Vector
9	ERT inverted models	Fieldwork (RES2DINV)	CSV / XYZ	Profile
10	GPR depth slices	Fieldwork (ReflexW)	GeoTIFF / XYZ	cm-scale
11	Borehole logs	Fieldwork	CSV / Excel	Point

PHASE 2 — ArcGIS Pro Project Setup & Data Import

2.1 Create a New ArcGIS Pro Project

1. Open ArcGIS Pro. Click **Map** under New Project.
2. Name: `Abidjan_Dam_Stability`. Location: choose a local folder (e.g., `C:\GIS\Abidjan_Dam\`).
3. A file geodatabase `Abidjan_Dam_Stability.gdb` is created automatically.
4. In the **Contents** pane, right-click the Map → **Properties › Coordinate Systems**. Set to **WGS 1984 UTM Zone 30N (EPSG:32630)**.

2.2 Set Geoprocessing Environments

Go to **Analysis › Environments** and set:

- bullet **Current Workspace** = `Abidjan_Dam_Stability.gdb`
- bullet **Output Coordinate System** = WGS 1984 UTM Zone 30N
- bullet **Processing Extent** = (leave blank for now; set after clipping boundary is ready)
- bullet **Cell Size** = 30 (we will standardise all rasters to 30 m to match the DEM)
- bullet **Mask** = (set later to Abidjan boundary polygon)

> Setting environments once avoids repetitive parameter entry for each tool.

2.3 Import Administrative Boundaries

1. Unzip the downloaded admin boundary shapefile.
2. In the **Catalog** pane, navigate to the folder, drag the shapefile onto the map.
3. The layer likely arrives in WGS84 (EPSG:4326). ArcGIS Pro projects it on-the-fly to UTM 30N for display.
4. Open the **Attribute Table**. Find the field that identifies communes (e.g., `ADM2_FR` or `admin2Name`).
5. **Select By Attributes**: build a query to select only communes inside the Abidjan district. Example:

```
ADM1_FR = 'ABIDJAN'
```

1 With the selection active → right-click layer → **Data › Export Features**. Output = `Abidjan_Communes` (in the .gdb).

2 Run **Data Management › Dissolve** on `Abidjan_Communes` (dissolve all) to create a single-polygon boundary: `Abidjan_Boundary`.

3 Now update environments: set **Processing Extent** and **Mask** to `Abidjan_Boundary`.

2.4 Import and Prepare the DEM

1. Drag the SRTM GeoTIFF(s) into the map.
2. If multiple tiles: **Data Management › Raster › Mosaic To New Raster**. Output = `DEM_Mosaic`.
3. **Spatial Analyst › Extraction › Extract by Mask**. Input = DEM_Mosaic, Mask = `Abidjan_Boundary`. Output = `DEM_Abidjan`.
4. **Spatial Analyst › Hydrology › Fill**. Input = DEM_Abidjan. Output = `DEM_Fill`. This removes depressions so water flows correctly.

2.5 Derive Slope

1. **Spatial Analyst › Surface › Slope**. Input = `DEM_Fill`. Output measurement = DEGREE. Output = `Slope_Abidjan`.

2 Slope is a key factor in both flood susceptibility and dam embankment stability.

2.6 Derive Hydrological Layers (Flow Direction, Accumulation, Watersheds)

1 **Spatial Analyst › Hydrology › Flow Direction**. Input = DEM_Fill. Output = `FlowDir`.

2 **Flow Accumulation**. Input = FlowDir. Output = `FlowAcc`.

3 To delineate a dam's upstream catchment: place a point (Snap Pour Point) at the dam location, then run **Watershed** tool with FlowDir + Pour Point. Output = `Dam_Catchment`.

4 To extract a stream network: **Raster Calculator**: `Con("FlowAcc" > 500, 1)` → `Streams_Raster`. Then **Stream to Feature** for a line feature class.

2.7 Import and Process Rainfall

1 If using **WorldClim BIO12** (easiest): drag the raster into the map, then **Extract by Mask** to Abidjan.

2 If using **CHIRPS monthly**: load all monthly rasters (e.g., 12 files for an average year), run **Cell Statistics** (Overlay type = MEAN) to create mean annual rainfall, then Extract by Mask. Output = `Rainfall_Abidjan`.

2.8 Import Land Use / Land Cover

1 Drag the BNETD / ESA WorldCover GeoTIFF into the map.

2 **Extract by Mask** to Abidjan boundary. Output = `LULC_Abidjan`.

3 Open **Symbology › Unique Values**. Assign meaningful colour ramps (green = forest, grey = built-up, blue = water, etc.).

2.9 Import Soil Data

1 For HWSD: drag the raster into the map, Extract by Mask. The raster cells contain a numeric Mapping Unit ID (MU_GLOBAL). Join the attribute database (.mdb or .csv shipped with HWSD) to extract texture class, drainage class.

2 For SoilGrids: download the clay / sand tiles, Extract by Mask. These are continuous 0–100% rasters.

3 Output = `Soil_Texture_Abidjan` or `Clay_Pct_Abidjan`.

2.10 Import Dam / Barrage Locations

1 Prepare your CSV with columns: `Dam_Name, Latitude, Longitude, Type, Height_m, Year_Built`.

2 In ArcGIS Pro: **Map › Add Data › XY Point Data**. Point to the CSV, X = Longitude, Y = Latitude, CRS = WGS84.

3 A temporary *Events* layer appears. Right-click → **Data › Export Features** to make it permanent: `Dam_Sites`.

2.11 Create a Study Area / Location Map

A study area map shows where your project is within the wider region. Every thesis and report needs one.

1 Insert a new **Layout**: Insert › New Layout › A4 Landscape.

2 Insert a **Map Frame** and drag it to fill most of the page.

3 Add the `Abidjan_Boundary` polygon (bold outline, no fill or light fill).

4 Add a **basemap** (Insert › Basemap › Topographic or Imagery).

5. Add the `Dam_Sites` points (red triangle symbols).
6. Add the river network lines (blue, thin).
7. Insert an **Inset Map** (Insert › Map Frame) showing the whole of Côte d'Ivoire with a red box around Abidjan to give geographic context.
8. Add map elements: **Scale Bar**, **North Arrow**, **Legend**, **Title** ("Study Area – Abidjan, Côte d'Ivoire"), and your name/date.
9. Export: **Share › Export Layout** as PDF (300 DPI) or PNG.

> Tip: Use a light basemap so your overlaid features stand out. The "Light Gray Canvas" basemap works well for academic maps.

PHASE 3 — Integrating Geophysical & Geotechnical Data into GIS

This is the most unique part of the project. We bring subsurface investigation results into ArcGIS Pro so they can be spatially combined with surface factors.

3.1 Import ERT (Electrical Resistivity Tomography) Data

What you have after inversion: a CSV/XYZ file with columns like `X_local`, `Depth`, `Resistivity_Ohm_m`. X_local is the distance along the profile.

Step A — Georeference the profile

1. You recorded the GPS coordinates of the first electrode (start) and last electrode (end) of each ERT line.
2. Open your CSV in Excel / LibreOffice. Add columns `Easting` and `Northing` by linearly interpolating between start and end coordinates based on `X_local`.
3. Formula (in Excel): `Easting = X_start + (X_local / Profile_Length) * (X_end - X_start)`. Same for Northing.
4. Add a `Elevation` column = Surface_Elevation − Depth (use DEM value at the point, or surveyed elevation).
5. Save the file as `ERT_Profile1_georef.csv`.

Step B — Import into ArcGIS Pro as 3D points

1. **Map › Add Data › XY Point Data**. X = Easting, Y = Northing, Z = Elevation. CRS = UTM 30N.
2. Export to geodatabase as `ERT_Profile1_Pts`.
3. Symbolise by `Resistivity_Ohm_m` using a **Graduated Colors** scheme (blue = low resistivity/clay/saturated, red = high resistivity/dry/granular).

Step C — Create an ERT resistivity surface (optional interpolation)

1. If you have multiple closely-spaced profiles forming a grid, you can interpolate a 3D resistivity volume.
2. **Geostatistical Analyst › Geostatistical Wizard**. Choose **Kriging/Co-Kriging** or **IDW**. Input = ERT points, attribute = Resistivity.
3. For a single profile, create a cross-section instead: in a **Local Scene**, drape the points and connect them visually.
4. For the multi-criteria model, you will extract a **surface resistivity map** (the shallowest model cells only, e.g., 0–2 m depth). Filter the CSV to `Depth <= 2`, then interpolate that subset to a raster: `ERT_Resistivity_Surface`.

> Why surface resistivity matters: low resistivity (<50 Ω·m) near the dam surface indicates saturated zones, clay lenses, or seepage paths — all indicators of potential weakness.

3.2 Import GPR (Ground Penetrating Radar) Data

What you have after processing: either depth-slice amplitude maps (GeoTIFF) or an XYZ point cloud (X, Y, Depth, Amplitude).

Option A — Georeferenced depth-slice rasters

1. If your GPR processing software (GPR-SLICE, ReflexW) exports georeferenced GeoTIFFs, simply drag them into ArcGIS Pro.
2. Each raster represents reflections at a specific depth (e.g., 0.5 m, 1.0 m, 1.5 m).
3. Symbolise with a stretched colour ramp. High-amplitude anomalies indicate interfaces, voids, or objects.

4 For the multi-criteria model, pick the most informative depth slice (typically 0.5–1.5 m for dam bodies) and rename it `GPR_Amplitude_Surface`.

Option B — XYZ point cloud

1 Import XYZ file as **XY Point Data** (X, Y, Z=Depth or Amplitude). Export to geodatabase.

2 Interpolate using **IDW** or **Natural Neighbor** to produce a continuous raster: `GPR_Amplitude_Surface`.

Option C — No GPR data yet (demo approach)

1 Create a dummy raster by rasterising a proximity-to-known-anomalies layer (you can refine later with real data).

2 This keeps the workflow running so you learn the method.

3.3 Import Geotechnical Borehole Data

1 Prepare CSV: `Borehole_ID, Easting, Northing, Depth_from_m, Depth_to_m, Lithology, SPT_N, Moisture_pct, Permeability_m_s`.

2 **Add XY Point Data** → Export to `Boreholes`.

3 For the vulnerability model, you want a single “weakness indicator” per borehole location. Create a field `Geotech_Score` based on the worst (weakest) layer in each borehole:

```
Score 3 (High weakness) = SPT_N < 5 or Permeability > 1e-4 m/s
Score 2 (Medium) = 5 ≤ SPT_N < 15 or 1e-6 < Permeability ≤ 1e-4
Score 1 (Low weakness) = SPT_N ≥ 15 and Permeability ≤ 1e-6
```

1 Interpolate `Geotech_Score` using **IDW** to create `Geotech_Weakness_Surface`. This is a very rough regionalization; its validity is limited to the immediate dam site.

> If you only have a few boreholes (common), treat the interpolated surface as indicative only. In your report, state the limitations.

PHASE 4 — Multi-Criteria Vulnerability Analysis (AHP + Weighted Overlay)

We use the **Analytic Hierarchy Process (AHP)** to assign scientifically justified weights to each factor, then combine them in a **Weighted Overlay** in ArcGIS Pro to produce a Dam Vulnerability Index map.

4.1 Define Vulnerability Criteria

Based on literature and engineering judgement, we use these factors:

#	Factor	Source Layer	Why it matters
1	Surface Resistivity (ERT)	ERT_Resistivity_Surface	Low resistivity = saturated / weak zones inside dam
2	GPR Amplitude Anomalies	GPR_Amplitude_Surface	Strong reflections = voids, cracks, material changes
3	Geotechnical Weakness	Geotech_Weakness_Surface	Low SPT / high permeability = structural risk
4	Slope of Dam Embankment	Slope_Abidjan (clipped near dam)	Steep slopes increase failure probability
5	Proximity to Drainage	Drainage_Proximity	Close to water = seepage and erosion risk
6	Rainfall Intensity	Rainfall_Abidjan	High rainfall increases hydraulic load
7	Land Use / Imperviousness	LULC_Abidjan	Built-up land increases run-off to dam
8	Soil Permeability / Clay %	Clay_Pct_Abidjan or Soil_Texture	Sandy soil = high seepage; clay = swelling risk

4.2 Reclassify All Factors to a Common 1–5 Scale

Each raster must be reclassified so that **1 = lowest vulnerability** and **5 = highest vulnerability**. Use **Spatial Analyst › Reclass › Reclassify**.

Example reclassification tables:

Slope

Slope (degrees)	Vulnerability class
0 – 5	1 (Very low)
5 – 15	2 (Low)
15 – 25	3 (Moderate)
25 – 35	4 (High)
> 35	5 (Very high)

Rainfall

Mean annual rainfall (mm)	Vulnerability class

< 1400	1
1400 – 1600	2
1600 – 1800	3
1800 – 2100	4
> 2100	5

ERT Surface Resistivity (inverted scale — low resistivity = high vulnerability)

Resistivity (Ω·m)	Vulnerability class
> 500	1 (dry, competent rock)
200 – 500	2
50 – 200	3
10 – 50	4 (saturated clay)
< 10	5 (highly saturated / seepage)

Repeat similar logic for each factor. For LULC: reclassify the categorical land-cover codes (e.g., Built-up = 5, Bare soil = 4, Cropland = 3, Shrub = 2, Forest = 1).

> Name outputs consistently: `Slope_Reclass`, `Rain_Reclass`, `ERT_Reclass`, `GPR_Reclass`, `Geotech_Reclass`, `Drainage_Reclass`, `LULC_Reclass`, `Soil_Reclass`.

4.3 Compute AHP Weights

The Analytic Hierarchy Process works by pairwise comparison of criteria. You build a square matrix, normalise it, and compute the priority vector (weights). Below is a suggested matrix:

	ERT	GPR	Geotech	Slope	Drainage	Rain	LULC	Soil
ERT	1	2	2	3	3	4	5	4
GPR	1/2	1	1	2	2	3	4	3
Geotech	1/2	1	1	2	2	3	4	3
Slope	1/3	1/2	1/2	1	1	2	3	2
Drainage	1/3	1/2	1/2	1	1	2	3	2
Rain	1/4	1/3	1/3	1/2	1/2	1	2	1
LULC	1/5	1/4	1/4	1/3	1/3	1/2	1	1/2
Soil	1/4	1/3	1/3	1/2	1/2	1	2	1

After normalizing and averaging rows, the approximate weights are:

Factor	Weight (approx.)
ERT (Surface Resistivity)	0.28
GPR (Amplitude Anomaly)	0.18
Geotechnical Weakness	0.18
Slope	0.11

Drainage Proximity	0.11
Rainfall	0.06
Land Use / Land Cover	0.03
Soil	0.05

Consistency Ratio (CR) must be < 0.10 for the matrix to be acceptable. Calculate it in Excel or use an online AHP calculator (e.g., `bpmsg.com/ahp-calculator`). If CR > 0.10, revise pairwise judgements.

These weights reflect that **direct subsurface measurements (ERT, GPR, geotechnical)** should dominate the model because they describe the dam body itself, while topographic and climatic factors provide context.

4.4 Run the Weighted Overlay in ArcGIS Pro

ArcGIS Pro has a dedicated **Weighted Overlay** tool that handles everything once your rasters are reclassified to the same scale.

1. Go to **Spatial Analyst › Overlay › Weighted Overlay**.
2. Add each reclassified raster. The tool shows each raster and its class values (1–5).
3. For each raster, set the **% Influence** = weight × 100. Round to whole percentages that sum to 100:

Raster	% Influence
ERT_Reclass	28
GPR_Reclass	18
Geotech_Reclass	18
Slope_Reclass	11
Drainage_Reclass	11
Rain_Reclass	6
LULC_Reclass	3
Soil_Reclass	5
Total	**100**

1. Leave all scale values as 1–5 (already done). Ensure Restricted is not set for any value.
2. Output raster = `Vulnerability_Index`.
3. The output has integer values 1–5 representing the vulnerability class of each 30 m pixel.

4.5 Alternative: Raster Calculator (Manual Weighted Sum)

If you prefer explicit control or the Weighted Overlay tool is unavailable:

```
("ERT_Reclass" * 0.28) + ("GPR_Reclass" * 0.18) + ("Geotech_Reclass" * 0.18) +
("Slope_Reclass" * 0.11) + ("Drainage_Reclass" * 0.11) + ("Rain_Reclass" * 0.06) +
("LULC_Reclass" * 0.03) + ("Soil_Reclass" * 0.05)
```

Output = `Vulnerability_Index_Continuous` (float). Reclassify to 5 classes using Natural Breaks or equal intervals.

PHASE 5 — Flood Risk Integration & Decision-Support Map Production

5.1 Extract Vulnerability per Dam Site

1 Buffer each dam point by its footprint radius (e.g., 200 m) using **Analysis › Buffer**: `Dam_Sites_Buffer`.

2 **Spatial Analyst › Zonal › Zonal Statistics as Table**. Input zones = Dam_Sites_Buffer, Zone field = Dam_Name, Value raster = Vulnerability_Index, Statistics = MEAN, MAX, MAJORITY.

3 Join the table back to Dam_Sites. Now each dam has a mean vulnerability score.

4 Classify dams: Mean ≥ 4 = **Critical**, 3–4 = **High**, 2–3 = **Moderate**, < 2 = **Low**.

5.2 Downstream Flood Inundation Scenario (Simplified)

A full dam-break simulation requires HEC-RAS. For a GIS-only approximation (suitable for a student project):

1 Identify the dam crest elevation from the DEM (use **Extract Values to Points** at the dam location).

2 Use **Raster Calculator** to select all cells downstream of the dam that are lower than the crest elevation: `Con("DEM_Fill" < Crest_Elevation, 1)`. This is a very rough maximum inundation extent.

3 Refine by limiting the flood zone to the **watershed** downstream of the dam: clip the inundation raster by the downstream sub-catchment polygon.

4 Convert to polygon: **Raster to Polygon**. Output = `Flood_Zone_DamX`.

> This is a simplified "bathtub" model. State clearly in your report that it does not account for flow dynamics, travel time, or channel roughness.

5.3 Assess Population & Infrastructure at Risk

1 Download **WorldPop** population raster for CIV (100 m). Extract by Mask to Abidjan. = `Pop_Abidjan`.

2 **Zonal Statistics as Table**: zones = Flood_Zone_DamX, value = Pop_Abidjan, statistic = SUM. This gives the population inside the potential flood zone.

3 Overlay `Roads` and `Buildings` with the flood zone using **Clip** or **Intersect** to identify infrastructure at risk.

4 Summarise: number of buildings, km of road inside the zone.

5.4 Produce the Dam Vulnerability Map (Main Output)

1 Create a new **Layout** (A3 Landscape recommended for detail).

2 Insert a **Map Frame**. Set the visible extent to the area around the dam(s) of interest.

3 Layers to display (bottom to top):

bullet Basemap (Light Gray Canvas)

bullet `Vulnerability_Index` raster — symbology: 5-class colour ramp (green → yellow → orange → red → dark red)

bullet `Flood_Zone_DamX` polygon — blue hatched fill

bullet `Dam_Sites` points — black triangle, labelled with name

bullet `ERT profile lines` — dashed white line showing survey locations

bullet `Boreholes` points — circle, labelled BH-1, BH-2…

bullet Roads and rivers for context

1. Add: Title, Legend, Scale Bar, North Arrow, Coordinate Grid (Grids tab in Map Frame Properties), Data Sources note, Author/Date.
2. Export as PDF (300 DPI): **Share › Export Layout**.

5.5 Additional Thematic Maps to Produce

For a complete thesis or report, produce these maps as well:

- **Study Area / Location Map** (done in Phase 2)
- **Geology / Soil Map** — soil texture classes symbolised categorically
- **Slope Map** — graduated colour
- **Rainfall Map** — graduated colour (blue ramp)
- **Land Use / Land Cover Map** — categorical colours
- **Drainage Density Map** — river network with proximity buffers
- **ERT Cross-Section** — export from RES2DINV or recreate in ArcGIS Pro Local Scene
- **GPR Depth-Slice Map** — amplitude at key depth
- **Reclassified Factor Maps** (one per criterion, 1–5 scale)
- **AHP Weights Diagram** (create outside GIS in Excel or PowerPoint)
- **Final Vulnerability Map** (the main deliverable)
- **Flood Inundation Map** with affected infrastructure

PHASE 6 — Resources, References & Further Learning

6.1 Key Academic References

bullet Kouamé et al. (2016). “Flood risk assessment and mapping in Abidjan district using multi-criteria analysis (AHP) and geoinformation techniques.” *Geoenvironmental Disasters*, 3:3. (doi:10.1186/s40677-016-0044-y)

bullet Loke, M.H. (2004). “Tutorial: 2-D and 3-D electrical imaging surveys.” (RES2DINV manual — freely available online).

bullet Daniels, D.J. (2004). *Ground Penetrating Radar*, 2nd ed., IET. (GPR theory and applications).

bullet Saaty, T.L. (1980). *The Analytic Hierarchy Process*. McGraw-Hill. (AHP foundational text).

bullet Foster, M., Fell, R., & Spannagle, M. (2000). “The statistics of embankment dam failures and accidents.” *Canadian Geotechnical Journal*, 37(5). (Dam failure modes).

bullet Sjodahl, P. et al. (2008). “Resistivity investigation and monitoring for detection of internal erosion and anomalous seepage in dams.” *Near Surface Geophysics*, 6. (ERT on dams).

6.2 Free Online Courses & Tutorials

bullet **Esri MOOC “Getting Started with ArcGIS Pro”** — learn.arcgis.com (4 hrs, free with ArcGIS account).

bullet **Esri “Spatial Analysis” course** — learn.arcgis.com/en/paths/spatial-analysis/

bullet **Esri “Imagery and Remote Sensing”** — learn.arcgis.com/en/paths/imagery/

bullet **YouTube: “ArcGIS Pro Weighted Overlay tutorial”** (many free videos, e.g., GISGeography, GeoDelta Labs).

bullet **RES2DINV Tutorial PDF** — included with the software, or search “RES2DINV tutorial Loke”.

bullet **GPRPy** (open-source GPR processing in Python): github.com/NSGeophysics/GPRPy

bullet **QGIS tutorials** (if you want a free GIS alternative): qgistutorials.com

6.3 Data Portal Quick Reference

Portal	URL	What to get
USGS EarthExplorer	earthexplorer.usgs.gov	SRTM DEM, Landsat imagery
HDX (Humanitarian Data Exchange)	data.humdata.org	Admin boundaries, OSM extracts, population
GADM	gadm.org	Admin boundaries (all levels)
WorldClim	worldclim.org	Climate normals (precip, temp)
CHIRPS	data.chc.ucsb.edu	Monthly/daily rainfall
ESA WorldCover	worldcover2021.esa.int	10 m land cover
BNETD Land Cover	Google Earth Engine	CIV-specific 10 m LULC
SoilGrids	soilgrids.org	250 m soil properties
FAO HWSD	gaez.fao.org	Harmonized World Soil Database
HydroSHEDS	hydrosheds.org	Rivers, basins
WorldPop	worldpop.org	100 m population grids

GRanD	globaldamwatch.org/grand	Global dam database
OpenStreetMap (HDX)	data.humdata.org	Roads, buildings, waterways

6.4 Beginner Tips & Common Mistakes

- **Always project data before analysis.** Geographic CRS (degrees) gives wrong distances. Use UTM Zone 30N for Abidjan.
- **Set environments early.** Cell size, snap raster, mask, and extent prevent misaligned outputs.
- **Check NoData.** If any input raster has NoData inside your study area, Weighted Overlay will produce NoData there. Fill gaps first (use Con or Nibble tools).
- **Document everything.** Use the Geoprocessing History (Analysis › History) to track every tool run.
- **Save intermediate outputs.** Name them logically (e.g., `Slope_Reclass`, not `Reclass_Slope_final2_v3`).
- **Back up your geodatabase** regularly (copy the .gdb folder).
- **AHP consistency:** always check the Consistency Ratio. A bad matrix means your weights are arbitrary.
- **When in doubt, read the Esri help page** for the tool you are using (press F1 with the tool open).

6.5 Suggested Project Timeline (12–16 Weeks)

Week	Activity
1–2	Literature review; define study objectives clearly
3–4	Download all remote-sensing / open data; set up ArcGIS Pro project; create study area map
5–7	Fieldwork: conduct ERT & GPR surveys, collect borehole data, GPS dam locations
8–9	Process geophysical data (RES2DINV, ReflexW); export to GIS-compatible formats
10–11	Import all data into GIS; derive layers (slope, hydrology, drainage proximity); reclassify
12–13	AHP weight calculation; Weighted Overlay; produce vulnerability map
14	Flood scenario modelling; population/infrastructure at risk analysis
15–16	Final map production; write-up; peer review

Appendix — Complete Workflow Summary (Quick Reference)

1. Download admin boundaries, clip to Abidjan → `Abidjan_Boundary`
2. Download SRTM DEM, mosaic, clip, fill → `DEM_Fill`
3. Derive Slope → `Slope_Abidjan`
4. Derive Flow Direction + Accumulation + Watershed → `FlowDir`, `Dam_Catchment`
5. Import rainfall, clip → `Rainfall_Abidjan`
6. Import LULC, clip → `LULC_Abidjan`
7. Import soil data, clip → `Soil_Texture_Abidjan`
8. Import dam locations (CSV → Points) → `Dam_Sites`
9. Import drainage network, buffer 50/100 m, rasterize → `Drainage_Proximity`
10. Import ERT (CSV → Points → IDW interpolation) → `ERT_Resistivity_Surface`
11. Import GPR (GeoTIFF / XYZ → interpolation) → `GPR_Amplitude_Surface`
12. Import geotechnical data (CSV → Points → IDW) → `Geotech_Weakness_Surface`
13. Reclassify ALL 8 factors to 1–5 scale
14. Compute AHP pairwise comparison matrix → weights (check CR < 0.10)
15. Run **Weighted Overlay** with % Influence from AHP → `Vulnerability_Index`
16. Extract vulnerability per dam (Zonal Statistics)
17. Create simplified flood inundation zone
18. Assess population & infrastructure at risk
19. Produce all thematic maps + final vulnerability map layout
20. Export to PDF; publish to ArcGIS Online for dashboard (optional)

> Generated by Kiro — Comprehensive GIS Guide for Dam Stability Assessment, Abidjan, Côte d’Ivoire. Adapt layer names and parameters to your specific project.